

a Form a local target

Three soft-SFA forces

SLOW
 $l_t(z_t - p_t)$

NONCOLLAPSE
 $-\frac{\lambda_v}{\lambda_s} g \odot (z_t - \mu)$

DECORRELATE
 $+\frac{\lambda_c}{2\lambda_s} A(z_t - \mu)$

$$r_t = r_t^{\text{slow}} + r_t^{\text{var}} + r_t^{\text{cov}}$$

$$\hat{z}_t = \text{sg}(z_t - r_t)$$

b Define the neuron state

Residual, then one lateral pass

activation residual

$$z_t - \hat{z}_t$$

$$\downarrow J_{\phi, t}^{\top}$$

base neuron state

$$b_t = J_{\phi, t}^{\top} (z_t - \hat{z}_t)$$

same-layer inhibition

$$b_t \text{ --- } s_t$$

$$s_t = b_t - \eta \kappa M b_t$$

c Update local synapses

The neuron state supplies both rules

$s_{t,j}$

FEEDFORWARD GRADIENT
 $\nabla_{W_{ji}} \tilde{L}_a \propto s_{t,j} x_{t,i}$

LATERAL CHANGE
 $\Delta L_{jk}^{\text{lat}} \propto -s_{t,j} s_{t,k}$

$x_{t,i} \text{ --- } W_{ji} \text{ --- } s_{t,j}$

LOCAL IN TIME

one detached predecessor p_t ; no graph through time

LOCAL IN SPACE

updates read endpoint states; no error crosses layers